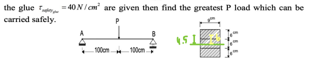


Problem: As seen in the figure, the beam with a span length of 2 meters is built from three equal timber parts (each have a rectangular cross section of 6cm*9cm) by means of gluing. If for the timber $\tau_{safety}=90N/cm^2$ $\sigma_{safety}=1300N/cm^2$ and for the glue $\tau_{safety_{glue}}=40N/cm^2$ are given then find the greatest P load which can be carried safely.

P
A
B
100cm
100cm
9cm
6cm
6cm
6cm
4.5

1. $R_A=R_B=P/2$, $T_{max}=P/2$, max. moment ortada: $M_{max}=R_A.100=50P$

2. $b=9cm$, $h=18cm$, $I_x=\frac{bh^3}{12}=4374cm^4$, $y_{max}=9cm$

3. $\sigma=\frac{My}{I_x}=\frac{(50P)\times(9)}{4374}\leq 1300 \rightarrow P\leq 12636N$

4. Ahsap kesilme: $\tau=\frac{TS}{I_xb}=\frac{(P/2)\times(9\times9)\times(4.5)}{4374\times9}\leq 90N \Rightarrow P\leq 19440N$

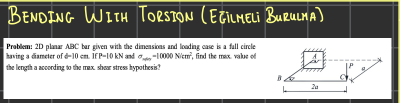
max. kesme nötr eksende olur, nötr eksenin üstündeki alan: $A'=9.9$
↳ $S=A.\bar{y}=(9\times9)\times(4.5)$ $\bar{y}=4.5cm$

5. Tutkal kesme: $\tau=\frac{TS}{I_xb}=\frac{(P/2)\times(9\times6)\times6}{4374\times9}\leq 40N \Rightarrow P\leq 9720N$

$A'=(9\times6)$, $\bar{y}=6cm \rightarrow S=(9\times6)\times6$

6. $P_{max}=9720N$

BENDING WITH TORSION (EĞILMELI BURULMA)



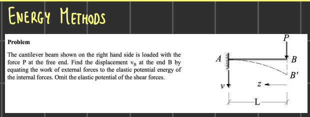
Problem: 2D planar ABC bar given with the dimensions and loading case is a full circle having a diameter of d=10 cm. If P=10 kN and σ_{safety}=10000 N/cm², find the max. value of the length a according to the max. shear stress hypothesis?

M_b: burulma momenti
M_e: eğilme momenti } $M_f=\sqrt{M_b^2+M_e^2}$ [max. shear stress hypothesis]

1. $M_b=P\times2a=2Pa$ $M_e=P\times a=Pa \Rightarrow M_f=2.236Pa$

2. Dolu Daire Kesit: $\sigma=\frac{M_f}{W_e}=\frac{32M_f}{\pi D^3}\leq\sigma_{all} \Rightarrow a\leq 43.906cm$

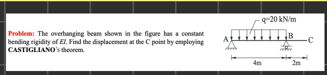
ENERGY METHODS



Problem

The cantilever beam shown on the right hand side is loaded with the force P at the free end. Find the displacement v_B at the end B by equating the work of external forces to the elastic potential energy of the internal forces. Omit the elastic potential of the shear forces.

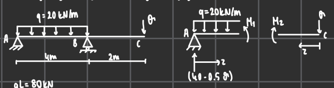
1. iç moment: $M=-Pz \rightarrow M^2=P^2z^2$
2. eğilmeden dolayı iç enerji: $U_i=\int_0^L\frac{M^2}{2EI}dz=\int_0^L\frac{P^2z^2}{2EI}dz=\frac{P^2L^3}{6EI}$
(kesme en. ihmal ediliyor)
3. dış kuvvetin yaptığı iş: $W_d=\frac{Pv_B}{2}$ (statik yükleme)
4. $\frac{P^2L^3}{6EI}=\frac{Pv_B}{2} \Rightarrow v_B=PL^3/3EI$



Problem: The overhanging beam shown in the figure has a constant bending rigidity of *EI*. Find the displacement at the C point by employing **CASTIGLIANO**'s theorem.

q=20 kN/m
A
B
C
4m
2m

1. C'de yük yok, hayali bir θ=0 yükü koyuyoruz



$qL=80kN$

$\Sigma M_B=0=A_y(4)-80(2)+\theta(2)=0 \Rightarrow A_y=40-0.5\theta$

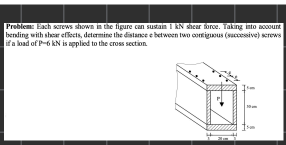
$\Sigma F_y=0=A_y+B_y-80-\theta \Rightarrow B_y=40+1.5\theta$

2. $0\leq z\leq4$; $M_1=A_y(z)-(qz)(z/2)=-10z^2+40z-0.5\theta z$, $\frac{\partial M_1}{\partial\theta}=-0.5z$

$0\leq z\leq2$; $M_2=-\theta z$, $\frac{\partial M_2}{\partial\theta}=-z$

3. Castigliano: $\delta_C=\frac{1}{EI}\int_0^4 M_1\frac{\partial M_1}{\partial\theta}dz+\frac{1}{EI}\int_0^2 M_2\frac{\partial M_2}{\partial\theta}dz \Rightarrow \theta=0$ alınır

$\delta_C=\frac{-106.67}{EI}$ → eksi işareti, yer değiştirmenin θ'nun tersi yönü olduğunu gösterir



Problem: Each screws shown in the figure can sustain 1 kN shear force. Taking into account bending with shear effects, determine the distance e between two contiguous (successive) screws if a load of P=6 kN is applied to the cross section.

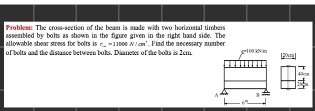
e
e
P
5 cm
30 cm
5 cm
3
20 cm
3

1. $I_x=\frac{26\times40^3}{12}-\frac{20\times30^3}{12}=93666.67cm^4$, $T=P=6kN$

2. üst levha: $S=A'\bar{y}=(26\times5)\times(17.5)$

3. $\tau=\frac{TS_x}{I_xb}=\frac{6\times(26\times5)\times(17.5)}{93666.67\times6}=0.024kN/cm^2$

4. $\tau\times e\times T=2\times1 \Rightarrow e=13.89cm$



Problem: The cross-section of the beam is made with two horizontal timbers assembled by bolts as shown in the figure given in the right hand side. The allowable shear stress for bolts is $\tau_{em}=11000\ N/cm^2$. Find the necessary number of bolts and the distance between bolts. Diameter of the bolts is 2cm.

q=100 kN/m
A
B
6m
20cm
40cm
20cm

1. $qL=600kN$, $R_A=R_B=300kN$, $T_{max}=300kN$

2. $I_x=\frac{bh^3}{12}=\frac{20\times60^3}{12}=360000cm^4$

3. $S_x=A'\bar{y}=(20\times20)\times(20)=8000cm^3$
(alt parça)

4. $z=\frac{S_xT_{max}}{I_x}=\frac{8000\times300000}{360000}=6666.67N/cm$
kesme akısı

5. Çivi: $A=\frac{\pi D^2}{4}$, $F_ç=A\tau_{all}=34557.52N$, $z.a\leq F_b \Rightarrow a\leq5.18cm \Rightarrow a=5cm$

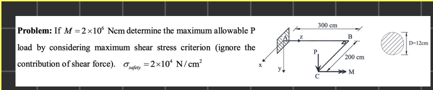
Deformation in Bending With Shear (162-167)

$(\Delta u_y)=k'_y\frac{T_y}{GA}\Delta z$
shear force
küçük kiriş elemanı uzunluğu
düşey kayma
shear modulus

$\Delta v_T=k'_y\frac{T_y}{GF}\Delta z \rightarrow \frac{dv_T}{dz}=k'_y\frac{T_y}{GF}$
shear'dan gelen yer değiştirme
alan (A)
shear force varsa slope değişimi olur

$\delta=\frac{PL^3}{3EI_x}+k'_y\frac{PL}{GF}=\frac{PL^3}{3EI_x}\left[1+\frac{3k'_yEI_x}{GFL^2}\right]$
normal bending
shear
$\delta=\delta_H+\delta_T$

$\frac{d^2v}{dz^2}=-\left[\frac{M_x}{EI_x}+\frac{k'_yP}{GF}\right]$
n.b.
s.



Problem: If $M=2\times10^6$ Ncm determine the maximum allowable P load by considering maximum shear stress criterion (ignore the contribution of shear force). $\sigma_{safety}=2\times10^4\ N/cm^2$

300 cm
A
z
B
P
200 cm
x
y
C
M
D=12cm

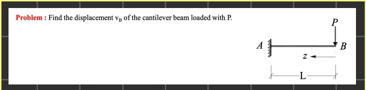
1. $\frac{M_f}{W_e}=\frac{32M_f}{\pi D^3}\leq\sigma_{safety}$, $I_0=\frac{\pi D^4}{32}=2034.72cm^4$, $I_x=\frac{\pi D^4}{64}=1017.36cm^4$
burulmada
eğilmede

2. $M_b=200P+2\times10^6 \rightarrow \tau=\frac{M_br}{I_0}=\frac{(200P+2\times10^6)\times6}{2034.72}=0.59P+5897.6$

$M_e=-300P \rightarrow \sigma=\frac{M_x}{I_x}y=\frac{(-300P)\times(-6)}{1017.36}=1.77P$

3. Max. Shear Stress C.: $\sigma_1-\sigma_2\leq\sigma_{safety}$, $\sigma_{1,2}=\frac{\sigma}{2}\pm\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$

$P=5118.12N=5.118kN$



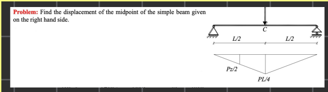
Problem : Find the displacement v_B of the cantilever beam loaded with P.

1. METOT: CASTIGLIANO TEOREMI

1. iç enerji: $U_i=\int_0^L\frac{M^2}{2EI}dz=\int_0^L\frac{(-Pz)^2}{2EI}dz=\frac{P^2L^3}{6EI}$
2. $v_B=\frac{\partial U_i}{\partial P}=\frac{2PL^3}{6EI}=\frac{PL^3}{3EI}$

2. METOT: CASTIGLIANO INTEGRALI

1. $\delta_K=\int_0^L\frac{M}{EI}\frac{\partial M}{\partial P_k}dz=\int_0^L\frac{(-Pz)}{EI}\frac{\partial(-Pz)}{\partial P}dz=\frac{PL^3}{3EI}$



Problem: Find the displacement of the midpoint of the simple beam given on the right hand side.

C
L/2
L/2
Pz/2
PL/4

$R_A=R_B=P/2$, $M_{max}=PL/4$

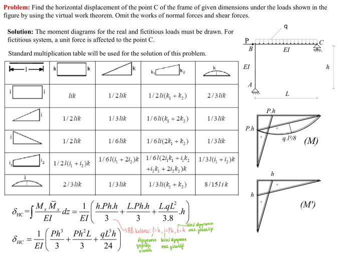
(U→δ)

1. $M=\frac{Pz}{2}$ (sol tarafta); sistem simetrik, sadece yarısı hesaplanıp 2 ile çarpılır

$U_i=2\left(\frac{1}{2}\int_0^{L/2}\frac{M^2}{2EI}dz\right)=\frac{1}{EI}\int_0^{L/2}\left(\frac{Pz}{2}\right)^2dz=\frac{P^2L^3}{96EI} \Rightarrow \delta_C=\frac{\partial U_i}{\partial P}=\frac{PL^3}{48EI}$ (castigliano)
diyagramın yarısı kullanılır

2. (Direkt Castigliano): $\delta_c=\int\frac{M}{EI}\frac{\partial M}{\partial P}dz=\frac{2}{EI}\int_0^{L/2}\left(\frac{Pz}{2}\right)\left(\frac{z}{2}\right)dz=\frac{PL^3}{48EI}$
simetriden
$M=Pz/2$ $\partial M/\partial P=z/2$

VIRTUAL WORK



Problem: Find the horizontal displacement of the point C of the frame of given dimensions under the loads shown in the figure by using the virtual work theorem. Omit the works of normal forces and shear forces.

Solution: The moment diagrams for the real and fictitious loads must be drawn. For fictitious system, a unit force is affected to the point C.

Standard multiplication table will be used for the solution of this problem.

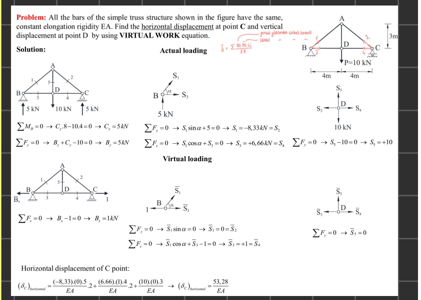
├─ l ─┤	k ▭ k	◢ k	k₁ ⏢ k₂	⌒ k
i ▭ i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
◢ i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i ◣	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i₁ ⏢ i₂	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2$ $+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
⌒ i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,lik$

$$\delta_{HC}=\int\frac{M_x\bar{M}_x}{EI}dz=\frac{1}{EI}\left(\frac{h.Ph.h}{3}+\frac{L.Ph.h}{3}+\frac{L.qL^2}{3.8}.h\right)$$

→AB kolonu: l=h, i=Ph, k=h
diyagramın yayıldığı uzunluk
birinci diyagramın max. yüksekliği
ikinci diyagramın max. yüksekliği

$$\delta_{HC}=\frac{1}{EI}\left(\frac{Ph^3}{3}+\frac{Ph^2L}{3}+\frac{qL^3h}{24}\right)$$

q
P
B
C
EI
EI
h
A
L
P.h
P.h
q.l²/8
(M)
h
h
(M')



Problem: All the bars of the simple truss structure shown in the figure have the same, constant elongation rigidity EA. Find the horizontal displacement at point **C** and vertical displacement at point D by using **VIRTUAL WORK** equation.

$\delta=\Sigma\frac{N_i\bar{N}_iL_i}{EA}$
gerçek yüklerden çubuk kuvveti
sanal

Solution:

Actual loading

$\sum M_B=0 \rightarrow C_y.8-10.4=0 \rightarrow C_y=5kN$

$\sum F_y=0 \rightarrow B_y+C_y-10=0 \rightarrow B_y=5kN$

$\sum F_y=0 \rightarrow S_1\sin\alpha+5=0 \rightarrow S_1=-8,33kN=S_2$

$\sum F_x=0 \rightarrow S_1\cos\alpha+S_3=0 \rightarrow S_3=+6,66kN=S_4$

$\sum F_y=0 \rightarrow S_5-10=0 \rightarrow S_5=+10$

Virtual loading

$\sum F_x=0 \rightarrow B_x-1=0 \rightarrow B_x=1kN$

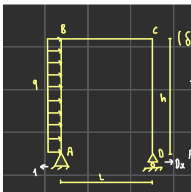
$\sum F_y=0 \rightarrow \bar{S}_1\sin\alpha=0 \rightarrow \bar{S}_1=0=\bar{S}_2$

$\sum F_x=0 \rightarrow \bar{S}_1\cos\alpha+\bar{S}_3-1=0 \rightarrow \bar{S}_3=+1=\bar{S}_4$

$\sum F_y=0 \rightarrow \bar{S}_5=0$

Horizontal displacement of C point:

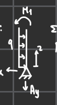
$$(\delta_C)_{horizontal}=\frac{(-8,33).(0).5}{EA}.2+\frac{(6.66).(1).4}{EA}.2+\frac{(10).(0).3}{EA} \rightarrow (\delta_C)_{horizontal}=\frac{53,28}{EA}$$



$(\delta_D)_{horizontal} = ?$

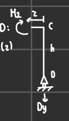
$\Sigma M_A = 0 = (qh)(h/2) - D_y(L) \Rightarrow \left[D_y = \frac{qh^2}{2L},\ A_y = \frac{-qh^2}{2L},\ A_x = qh \right]$

ACTUAL LOADING



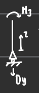
AB: $\Sigma M = 0 = M_1 + qz(z/2) - A_x(z)$

$M_1 = qhz - \frac{qz^2}{2}$



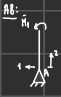
CD: $\Sigma M = 0 = M_2 - D_y(z)$

$M_2 = \frac{qh^2}{2L} z$



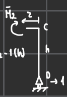
$\Sigma M = 0 = M_3$

VIRTUAL LOADING



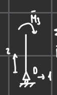
AB: $\Sigma M = \bar{M}_1 - 1(z) = 0$

$\bar{M}_1 = z$



CD: $\Sigma M = 0 = \bar{M}_2 - 1(h)$

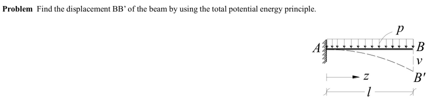
$\bar{M}_2 = h$



$\Sigma M = 0 = \bar{M}_3 - 1(z)$

$\bar{M}_3 = z$

$$(\delta_D)_{horizontal} = \Sigma \int \frac{M_i \bar{M}_i}{EI} dz = \frac{1}{EI}\left(\int_0^h \left(qhz - \frac{qz^2}{2}\right)(z)dz + \int_0^L \left(\frac{qh^2}{2L} z\right)(h)\,dz + \int_0^h (0)(z)\,dz \right] = \frac{5qh^4}{24EI} + \frac{qh^3L}{4EI}$$


Problem Find the displacement BB’ of the beam by using the total potential energy principle.

1. A noktasında : $v(0) = 0$ (deplasman) ; $v'(0) = 0$ (dönme) } v(z) fonksiyonu en az "z^2" ile başlamalı

$\hookrightarrow v(z) = a_2z^2 + a_3z^3 + a_4z^4$; $v(0) = 0$

$v'(z) = 2a_2z + 3a_3z^2 + 4a_4z^3$; $v'(0) = 0$ } sınır şartlarını sağlar

2. eğilmeden gelen iç enerji : $U = \int_0^L \frac{1}{2} EI_x (v'')^2 dz$

yayılı yükün yaptığı dış iş: $W = \int_0^L p v(z) dz$

} toplam potansiyel enerji : $\Pi = U - W$

$$\Pi = \int_0^L \left[\frac{1}{2} EI_x (v'')^2 - pv \right] dz$$

3. $v(z) = a_2z^2 + a_3z^3 + a_4z^4$

$v'(z) = 2a_2z + 3a_3z^2 + 4a_4z^3$

$v''(z) = 2a_2 + 6a_3z + 12a_4z^2 \rightarrow M = EI_x v''$

4. min. enerji şartı : $\frac{\partial \Pi}{\partial a_2}^{(a)} = 0,\ \frac{\partial \Pi}{\partial a_3}^{(b)} = 0,\ \frac{\partial \Pi}{\partial a_4}^{(c)} = 0$

(a) → $\frac{EI_x}{2}(8a_2 l + 12a_3 l^2 + 16a_4 l^3) - \frac{pl^3}{3} = 0$

(b) → $\frac{EI_x}{2}(12a_2 l^2 + 24a_3 l^3 + 36a_4 l^4) - \frac{pl^4}{4} = 0$

(c) → $\frac{EI_x}{2}(16a_2 l^3 + 36a_3 l^4 + \frac{288}{5} a_4 l^5) - \frac{pl^5}{5} = 0$

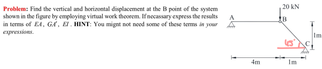
} $a_2 = \frac{pl^2}{4EI_x}$; $a_3 = \frac{-pl}{6EI_x}$; $a_4 = \frac{p}{24EI_x}$

5. a_2, a_3, a_4 ; v(z) denkleminde yerine konulur

$$v(z) = \frac{pl^4}{24EI_x}\left[6\left(\frac{z}{l}\right)^2 - 4\left(\frac{z}{l}\right)^3 + \left(\frac{z}{l}\right)^4 \right]$$

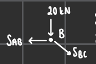
B noktasında $z = l$; $v(l) = \frac{pl^4}{8EI_x}$

Kolon Uç Durumu	Etkin Boy (l_B)
Mafsal - Mafsal	L
Ankastre - Mafsal	0.7L
Ankastre - Ankastre	0.5L
Ankastre - Serbest	2L


Problem: Find the vertical and horizontal displacement at the B point of the system shown in the figure by employing virtual work theorem. If necassary express the results in terms of EA, GA', EI. **HINT**: You mignt not need some of these terms *in your expressions.*

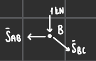
$$\boxed{\delta = \Sigma \frac{N_i \bar{N}_i L_i}{EA}}$$

1. Düşey Yer Değiştirme



$S_{AB} = -20 kN$

$S_{BC} = -20\sqrt{2}$ kN

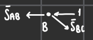


$\bar{S}_{AB} = -1$ kN

$\bar{S}_{BC} = -\sqrt{2}$ kN

$$(\delta_B)_{vertical} = \frac{(-20)(-1)(4)}{EA} + \frac{(-20\sqrt{2})(-\sqrt{2})(\sqrt{2})}{EA} = \frac{136.57}{EA}$$

2. Yatay Yer Değiştirme



$\bar{S}_{AB} = -1$, $\bar{S}_{BC} = 0$

$$(\delta_B)_{horizontal} = \frac{(-20)(-1)(4)}{EA} + \frac{(-20\sqrt{2})(0)(\sqrt{2})}{EA} = \frac{80}{EA}$$

ELASTIC STABILITY - BUCKING

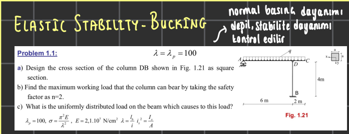
normal basınç dayanımı değil, stabilite dayanımı kontrol edilir

$\lambda = \frac{l_b}{i}$ — burkulma boyu ; atalet yarıçapı ($i^2 = I/A$) ; narinlik oranı (slenderness ratio)

Verilenler: $\lambda = 100$, $E = 2.1 \times 10^7 N/cm^2$

$l = 400cm$, $l_b = 2l = 800cm$

altı ankastre, üstü serbest gibi davranıyor


Problem 1.1: $\lambda = \lambda_p = 100$

a) Design the cross section of the column DB shown in Fig. 1.21 as square section.

b) Find the maximum working load that the column can bear by taking the safety factor as n=2.

c) What is the uniformly distributed load on the beam which causes to this load?

$\lambda_p = 100,\ \sigma = \frac{\pi^2 E}{\lambda^2},\ E = 2{,}1.10^7\ N/cm^2\ \ \lambda = \frac{l_b}{i}\ \ i_x^2 = \frac{I_x}{A}$

Fig. 1.21

1. kare kesit için: $I_x = \frac{a.a^3}{12}$, $A = a^2 \Rightarrow i = \sqrt{\frac{a^4}{12} \cdot \frac{1}{a^2}} = a/2\sqrt{3}$

$\lambda = \frac{l_b}{i} = \frac{800}{a/2\sqrt{3}} = 100 \Rightarrow a = 27.71$ cm

2. Euler kritik gerilmesi : $\sigma_k = \frac{\pi^2 E}{\lambda^2} = 20\,726.17 N/cm^2$

kritik burulma yükü : $P_k = \sigma_k A = 15\,914.47$ kN

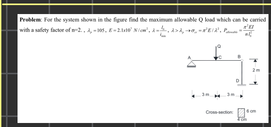
emniyet katsayısı : $P_{em} = P_k/n = 7957.24$ kN

3. DB'nin taşıdığı düşey kuvvet : $P_{em} = 7957.24$ kN

$\Sigma M_A = 7957.24(6) - 8q(4) = 0 \Rightarrow q = 1491.98$ kN/m

1. l_b bul 2. kesitten I, A, i bul 3. $\lambda = l_b/i$ bul

4. Euler gerilmesi : $\sigma_k = \pi^2 E/\lambda^2$ 5. Kritik yük : $P_k = \sigma_k A$ 6. Emniyet k. : $P_{em} = P_k/n$

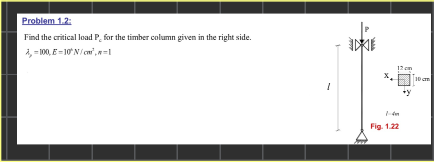

Problem: For the system shown in the figure find the maximum allowable Q load which can be carried with a safety factor of n=2., $\lambda_p = 105$, $E = 2.1 \times 10^7\ N/cm^2$, $\lambda = \frac{l_b}{i_{min}}$, $\lambda > \lambda_p \rightarrow \sigma_{cr} = \pi^2 E/\lambda^2$, $P_{allowable} = \frac{\pi^2 EI}{n l_b^2}$

1. $l = 200cm$, $l_b = 0.7l = 140cm$, $I_{min} = \frac{6 \times 4^3}{12} = 32cm^4$ (küçük I_x alınır), $A = 24cm^2$, $i_{min} = \sqrt{\frac{I_{min}}{A}} = 1.155cm$

2. $\lambda = \frac{l_b}{i_{min}} = 121$, $\lambda_p = 105 \Rightarrow \lambda > \lambda_p$: euler burkulma kullanılır

3. $P_{all} = \frac{\pi^2 EI_{min}}{n l_b^2} = 169\,021.7 N$ BD kolonunun taşıyabileceği max. basınç kuvveti

4. $A_y = B_y = Q/2$, BD kolonu B mesnet tepkisini taşır $\Rightarrow \frac{Q}{2} \leq 169\,021.7 \rightarrow Q \leq 338\,043.4 N \Rightarrow Q_{max} = 338.04$ kN


Problem 1.2:

Find the critical load P_c for the timber column given in the right side.

$\lambda_p = 100, E = 10^6 N/cm^2, n = 1$

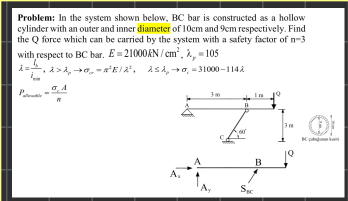
Fig. 1.22

$l_b = l = 400cm$, $I_x = \frac{12 \times 10^3}{12} = 1000 cm^4$, $I_y = \frac{10 \times 12^3}{12} = 1440 cm^4$

$I_{min} = 1000 cm^4 \rightarrow i = \sqrt{\frac{1000}{12 \times 10}} = 2.89 cm \rightarrow \lambda = l_b/i_{min} = 138.41$

$\lambda_p = 100 \Rightarrow \lambda > \lambda_p$: euler kullanılır ($\lambda < \lambda_p$: tetmajer / inelastik)

$P_k = \frac{\pi^2 EI}{n l_b^2} = 616\,850 N$


Problem: In the system shown below, BC bar is constructed as a hollow cylinder with an outer and inner diameter of 10cm and 9cm respectively. Find the Q force which can be carried by the system with a safety factor of n=3 with respect to BC bar. $E = 21000 kN/cm^2$, $\lambda_p = 105$

$\lambda = \frac{l_b}{i_{min}}$, $\lambda > \lambda_p \rightarrow \sigma_{cr} = \pi^2 E/\lambda^2$, $\lambda \leq \lambda_p \rightarrow \sigma_c = 31000 - 114\lambda$

$P_{allowable} = \frac{\sigma_c A}{n}$

1. $\Sigma M_A = 0 = S_{BC} \sin(60)(3) - Q(4) \rightarrow S_{BC} = 1.54Q$

2. $A = \frac{\pi}{4}(10^2 - 9^2) = 14.915 cm^2$, $I = \frac{\pi}{64}(10^4 - 9^4) = 168.73 cm^4 \rightarrow i = 3.36cm$

3. $l_b = \frac{300}{\sin(60)} = 346.4cm$, $\lambda = l_b/i = 103 \Rightarrow \lambda < \lambda_p$: inelastik burkulma formülü

4. $\lambda < \lambda_p \Rightarrow \sigma_c = 31000 - 114\lambda = 19\,230 N/cm^2$

5. $n = 3$, $P_{all} = \frac{\sigma_c A}{n} = 95.61 kN \rightarrow S_{BC} = 1.54Q = 95.61 \Rightarrow Q = 62.1 kN$